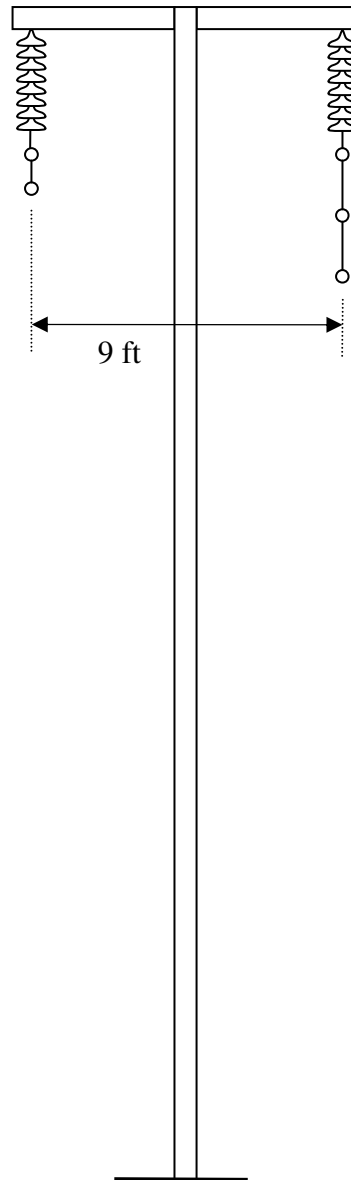


A single phase transmission line uses 2 ACSR “Waxwing” cables with 5” spacing in one conductor and three ACSR “Pelican” cables with 10” spacing between adjacent cables in the other conductor as shown in the support tower diagram below. The topmost cables of the transmission line are both at the same height. Calculate the propagation constant and characteristic impedance for this transmission line while operating at 30°C.



Code Name	Strands (Al/St)	Al Area (cmil)	diameter (in)	radius (ft)	GMR (ft)	R (Ω/mile) 60Hz, 20°C	ΔR/ΔT	Rated Amps
Waxwing	18/1	266800	0.609	0.0254	0.0198	0.3488	0.00141	465
Pelican	18/1	477000	0.814	0.0339	0.0264	0.1957	0.00079	670

$$GMD_{xy} = \left[\prod_{i=1}^2 \left(\prod_{j=1}^3 D_{xij} \right) \right]^{\frac{1}{6}} = \left[(D_{x1y1} D_{x1y2} D_{x1y3}) (D_{x2y1} D_{x2y2} D_{x2y3}) \right]^{\frac{1}{6}}$$

$$= \left\{ \left[(9) \left(\sqrt{9^2 + \left(\frac{10}{12} \right)^2} \right) \left(\sqrt{9^2 + \left(\frac{20}{12} \right)^2} \right) \right] \left[\left(\sqrt{9^2 + \left(\frac{5}{12} \right)^2} \right) \left(\sqrt{9^2 + \left(\frac{10-5}{12} \right)^2} \right) \left(\sqrt{9^2 + \left(\frac{20-5}{12} \right)^2} \right) \right] \right\}^{\frac{1}{6}} = 9.049368 \text{ ft}$$

$$GMR_x = \sqrt{d \cdot GMR_{xx}} = \sqrt{\left(\frac{5}{12} \right) (0.0198)} = 0.090830 \text{ ft}, \quad mGMR_x = \sqrt{d \cdot r_{sx}} = \sqrt{\left(\frac{5}{12} \right) (0.0254)} = 0.102875 \text{ ft}$$

$$GMR_y = \left[\prod_{i=1}^3 \left(\prod_{j=1}^3 D_{ij} \right) \right]^{\frac{1}{9}} = \left[(D_{11}D_{12}D_{13})(D_{21}D_{22}D_{23})(D_{31}D_{32}D_{33}) \right]^{\frac{1}{9}}$$

$$D_{11} = D_{22} = D_{33} = GMR_{sy} = 0.0264 \text{ ft}, \quad D_{12} = D_{21} = D_{23} = D_{32} = \left(\frac{10}{12} \right) \text{ ft}, \quad D_{13} = D_{31} = \left(\frac{20}{12} \right) \text{ ft}$$

$$GMR_y = \left[(0.0264)^3 \left(\frac{10}{12} \right)^4 \left(\frac{20}{12} \right)^2 \right]^{\frac{1}{9}} = 0.307593 \text{ ft}, \quad mGMR_y = \left[(0.0339)^3 \left(\frac{10}{12} \right)^4 \left(\frac{20}{12} \right)^2 \right]^{\frac{1}{9}} = 0.334329 \text{ ft}$$

$$R = \frac{R_{sx}}{n_x} + \frac{R_{sy}}{n_y} = \frac{(0.3488 \Omega / \text{mi}) + (0.00141)(30 - 20)}{2} + \frac{(0.1957 \Omega / \text{mi}) + (0.00079)(30 - 20)}{3} = 0.249317 \Omega / \text{mi}$$

$$L = k_{L1} \cdot \ln \left(\frac{GMD_{xy}}{\sqrt{GMR_x \cdot GMR_y}} \right) = 2.5695 \times 10^{-3} \text{ H} / \text{mi}$$

$$\vec{\mathcal{Z}} = R + j\omega L = 1.00026 \angle 75.567^\circ \Omega / \text{mi}$$

$$G = 0$$

$$C = k_{C1} \left/ \ln \left(\frac{GMD_{xy}}{\sqrt{mGMR_x \cdot mGMR_y}} \right) \right. = 1.1515 \times 10^{-8} \text{ H} / \text{mi}$$

$$\bar{y} = G + j\omega C = 4.34104 \times 10^{-6} \angle 90^\circ \text{ S} / \text{mi}$$

$$\bar{\gamma} = \sqrt{\vec{\mathcal{Z}} \cdot \bar{y}} = 2.08378 \times 10^{-3} \angle 82.783^\circ / \text{mi}$$

$$\bar{Z}_c = \sqrt{\vec{\mathcal{Z}} / \bar{y}} = 480.0196 \angle -7.217^\circ \Omega$$